

**Roads Are
No Fun**

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Design Verification Methods

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1 Overview

Design verification is the process of confirming that the design meets the requirements and was built correctly according to its specifications. To accomplish verification, one or more verification methods are used, chosen based on the requirement type, risk, cost, and feasibility. The basic verification methods are: analysis, test, inspection, demonstration, similarity, and lot controls. Typical requirement types include: functional, feature, performance, interface, environmental, and design requirements. The requirements should be provided in an item or system specification which also identifies the proper verification method. A requirements verification matrix (RVM) may be used to summarize the requirement information including: requirement identification, origin, text, phase of verification, verification method, pass / fail criteria, etc. A design verification plan and report (DVP&R) is a document which provides details on the verification plan and results.

1.1 Qualification versus Acceptance

Qualification tests establish margin for the design and may consume the test article (static load tests to failure or life tests). Acceptance tests ensure the end-item was built correctly with no flaws / defects and meets all performance requirements (workmanship). It is expected the qualification unit(s) will see all tests to a higher level or longer duration to establish margin. This also provides opportunity to refine and validate the verification plan. Delivered items undergo acceptance testing, which can be a reduced set of verification events to ensure quality.

1.2 Verification versus Validation

Verification checks whether a product meets its specified requirements and is “built right”. Validation checks whether the product fulfills its intended purpose and ensures the “right product is built”.

2 Verification Methods

2.1 Analysis

Analysis is verification through mathematical models, simulations, calculations, or theoretical evaluation. Analysis is typically fast and cost-effective, and it supports early design decisions before prototypes exist. Analysis is used when physical testing is impractical, expensive, or unnecessary and early in development to predict performance before hardware is available.

Examples:

- finite element analysis (FEA)
- computational fluid dynamics (CFD)
- tolerance stack-ups

- stress/strain calculations
- dynamic simulations
- kinematic analysis

2.2 Demonstration

Demonstration is verification by showing that the product performs a function under normal operating conditions without detailed measurement. Data may be observed which is available from the test item itself. Demonstration is used for qualitative or functional requirements and when special equipment or precise measurement is not necessary.

Examples:

- powering on a device to show it boots
- moving a mechanism through its full range of motion
- showing a user interface responds to input

2.3 Inspection

Inspection is verification by examining, measuring, or reviewing the physical item or documentation. Inspection is used for simple, observable requirements and does not require specialized measurement equipment or measurement while operating.

Examples:

- existence of a feature
- dimensional inspection
- visual inspection
- reviewing drawings, BOMs, or documentation
- critical clearance checks between moving parts at extreme ranges of motion

2.4 Test

Test is a verification by collection and evaluation of quantitative performance, functional, or environmental data. Testing requires specialized instrumentation or equipment and typically requires measurement during operation or application of external stimuli. Test is used when quantitative, measurable evidence is required or when existing models or knowledge is not mature enough or practical to predict the outcome.

Examples:

- static load testing (externally applied load)
- environmental testing (thermal, vibration, vacuum, humidity)
- performance testing (dynamometer)

2.5 Similarity

Verification by similarity is accomplished by showing that a requirement is met through comparison to a previously verified, equivalent design. Verification by similarity is only applicable to qualification level requirements, not acceptance level (workmanship). Verification by similarity must be used carefully with the following restrictions:

- heritage unit was not verified by similarity or analysis
- heritage unit was flight quality
- heritage unit environments envelope those for the candidate
- relevant design features are identical
- heritage unit was successfully qualified without waivers
- heritage and candidate functions and operation are identical
- heritage unit life envelopes that for the candidate
- all documentation for the heritage unit is available

Qualification by similarity can reduce time and cost but requires strong justification and documentation.

Examples:

- reusing a previously qualified material or component
- leveraging historical test data

2.6 Lot Controls

Lot controls is a verification by testing of samples from a well-controlled production process. Some products cannot be tested without consumption of the product; for example: explosive devices. The presumption is that the samples that are tested are representative of the rest of the delivered lot. Statistics can be used to bound the expected performance of the rest of the lot based on the test data of the tested samples. The number of samples to be tested is determined by the size of the lot and level of desired reliability. Non-destructive techniques can be used on the whole lot to detect defects that do not require consumption and confirm the representative assumption.

Examples:

- lot sample testing of explosive devices
- co-mixed and cured composite / epoxy samples subjected to tensile testing

3 Planning & Documentation

3.1 Design Specification

A design specification is a document which provides the details of design features, functions, requirements, constraints, and performance of an item. It provides information for proper usage of the item, including key requirements, verification methods, and expected performance (acceptance criteria). A specification may include a requirements verification matrix (RVM).

3.2 Requirements Verification Matrix (RVM)

A requirements verification matrix (RVM) is a table that summarizes each requirement, the method(s) used to verify it, and the verification plan or evidence. It may also include other planning information such as schedule, responsibility, traceability, etc. It helps ensure complete requirement coverage, traceability, and accountability across the verification process. The RVM is useful because it:

- Ensures no requirement is left unverified
- Provides a single reference for design teams, test teams, and quality
- Supports audits, design reviews, and regulatory compliance
- Enables early detection of verification gaps or ambiguous requirements

The RVM often contains:

- Requirement identification (ID)
- Requirement description (text)
- Source / origin (parent)
- Verification method
- Acceptance criteria
- Link to verification activity details (test plan or procedure)
- Responsibility
- Timing / Schedule
- High level results summary (pass/fail)

3.3 Design Verification Plan and Report (DVP&R)

A Design Verification Plan and Report (DVP&R) is a part of the engineering documentation used to plan, execute, and record all verification activities required to prove that a design meets its specifications. A typical DVP&R includes:

- List of requirements to be verified
- Verification method (analysis, test, inspection, demonstration, similarity, etc.)
- Test conditions and procedures
- Acceptance criteria
- Responsibility and timing
- Actual results and pass/fail status

It combines two parts:

- DVP — Design Verification Plan (the planned tests) = Test Plan
- DVR — Design Verification Report (the results) = Test Report

3.4 Verification (Test) Procedures

Verification (test) procedures are the detailed steps to be followed in the execution of the verification event. A test procedure may be interactive with test results added to the document as the test is executed.

Sloppy Copy = copy of the as-run test procedures which is marked up or modified during the verification as part of the procedure refinement.

References

1. SMC-S-016 Test Requirements for Launch, Upper-Stage and Space Vehicles, 2014.
2. SMC Systems Engineering Primer & Handbook, Space & Missile Systems Center U.S. Air Force, 2005.